IGCSE 0607 Practice Questions — E4.1

Coordinates

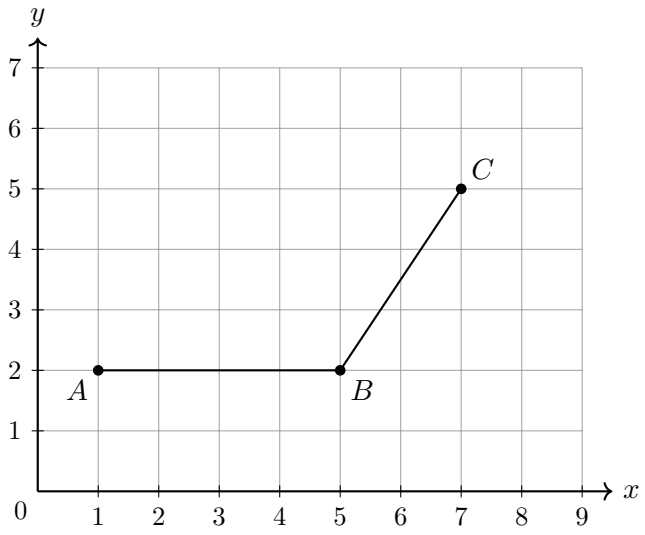
Total Questions: 6 — **Total Marks:** 39 Marks

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1 The diagram shows three vertices, $A(1, 2)$, $B(5, 2)$ and $C(7, 5)$, of a parallelogram $ABCD$.



(a) Find the coordinates of vertex D.

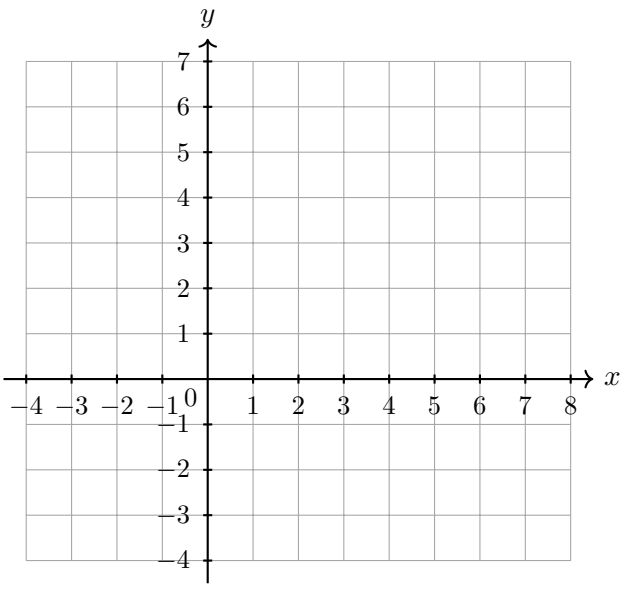
((..... , ...)) **[2]**

(b) Calculate the area of the parallelogram $ABCD$.

..... **[2]**

2 The points $A(-1, 3)$, $B(2, -1)$ and $C(6, 1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

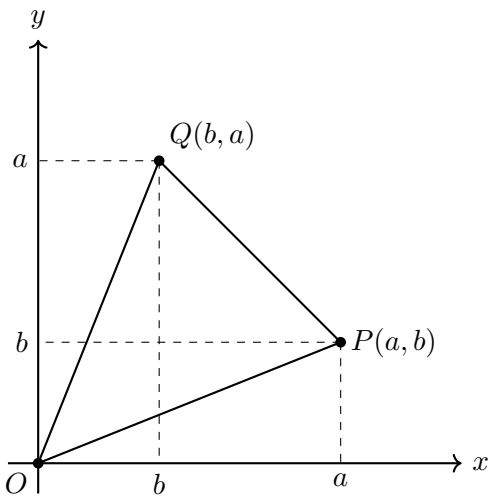
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

3 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

4 A rectangle has its sides parallel to the coordinate axes.
Two opposite vertices of the rectangle are $A(-2, -4)$ and $C(x, y)$, where $x > 0$ and $y > 0$.
The area of the rectangle is 54 square units and its perimeter is 30 units.

(a) Show that $x + y = 9$.

[2]

(b) Form an equation in x and solve it to find the two possible pairs of coordinates for C.

[4]

(c) For the case where $x > y$, write down the coordinates of the other two vertices of the rectangle.

[2]

5 A straight line has the equation $2y + 3x = 12$.
The line intersects the y-axis at A and the x-axis at B.
The point O is the origin $(0, 0)$.

(a) Find the coordinates of A and B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Point C is such that $OACB$ is a rectangle.
Write down the coordinates of C.

((..... , ...)) **[1]**

(c) Point P lies on the line segment AB and has an x-coordinate of 2.
Find the coordinates of P.

((..... , ...)) **[2]**

(d) Find the area of triangle OPB.

..... **[2]**

6 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]